CSC432 – Information Security

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Lecture No 5

Advanced Encryption Standard

Origins of AES

- Because of the exhaustive key search and theoretical attacks to break it, clearly a replacement for DES was needed
  - Triple-DES can be used but its slow and has small blocks
- NIST issued call for ciphers in 1997, 22 submissions, 15 candidates accepted in Jun 1998
- 5 were shortlisted in Aug 1999, MARS, RC6, Rijndael, Serpent, Twofish
- **Winner: Rijndael,** two Belgian cryptographers Joan Daemen and Vincent Rijmen
- A slight variation of the Rijndael cipher was selected as AES in Oct 2000
- Published by NIST as FIPS PUB 197 standard in Nov 2001

Rijndael Cipher

- It is an iterative cipher (operates on entire data block in every round) rather than Feistel (operate on halves at a time)
- It was designed to have characteristics of:
  - Resistance against all known attacks
  - Speed and code compactness on a wide range of platforms
  - Design simplicity
- Rijndael allows many block sizes and key sizes

Advanced Encryption Standard

- AES is a block cipher with a block length of 128 bits
- Allows for three different key lengths: 128, 192, or 256 bits
- Encryption consists of 10 rounds of processing for 128-bit keys, 12 rounds for 192-bit keys, and 14 rounds for 256-bit keys

Nr (number of rounds) = 6 + max{Nb, Nk}

Nb = 32-bit words in the block   Nk = 32-bit words in key

For AES-128 with 128-bit key: Nb = 4, Nk = 4, Nr = 10

Matrix = 4 rows × Nb columns

Advanced Encryption Standard

- Except for the last round, all other rounds are identical
- Each round of processing includes one single-byte based <u>substitution step</u>, a row-wise <u>permutation step</u>, a column-wise <u>mixing step</u>, and the <u>addition of the round key</u>

- The order in which these four steps are executed is different for encryption and decryption
- So, unlike DES, the decryption algorithm differs substantially from the encryption algorithm

Advanced Encryption Standard

- The input to the AES is a single 128-bit block, consisting of a 4×4 matrix of bytes, arranged as follows:

$$\begin{bmatrix} byte_0 & byte_4 & byte_8 & byte_{12} \\ byte_1 & byte_5 & byte_9 & byte_{13} \\ byte_2 & byte_6 & byte_{10} & byte_{14} \\ byte_3 & byte_7 & byte_{11} & byte_{15} \end{bmatrix}$$

- This 4 × 4 matrix of bytes is referred to as the **state array**
- Each column of the state array is a **word**, as is each row
  - A word consists of four bytes, that is 32 bits

Advanced Encryption Standard

- 128-bit key is also arranged in the form of matrix of 4 × 4 bytes
  - As with the input block, the first word from the key fills the first column of the matrix, and so on

- <u>Four column words of the key matrix are expanded into a schedule of **44 words**</u>

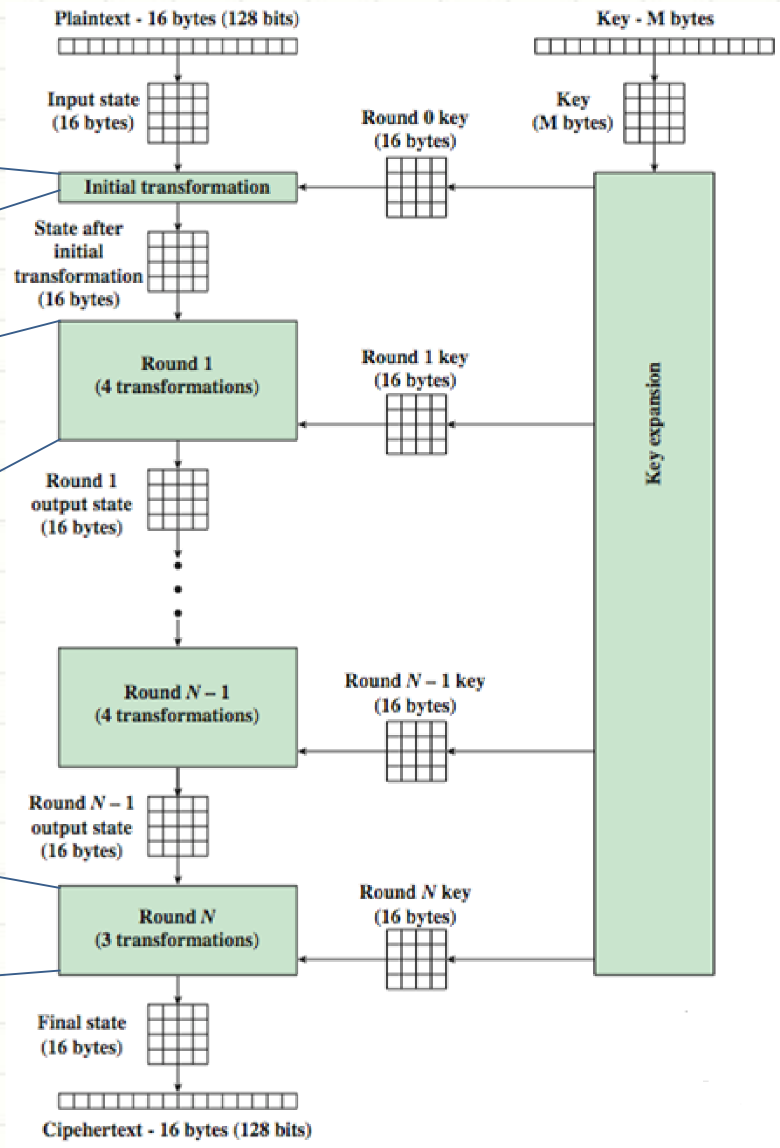
Advanced Encryption Standard

AES Encryption Process

The input state array is XORed with the first four words of the key schedule

1) Substitute bytes
2) Shift rows
3) Mix columns
4) Add round key

The last round for encryption does not involve the "Mix columns" step



Advanced Encryption Standard

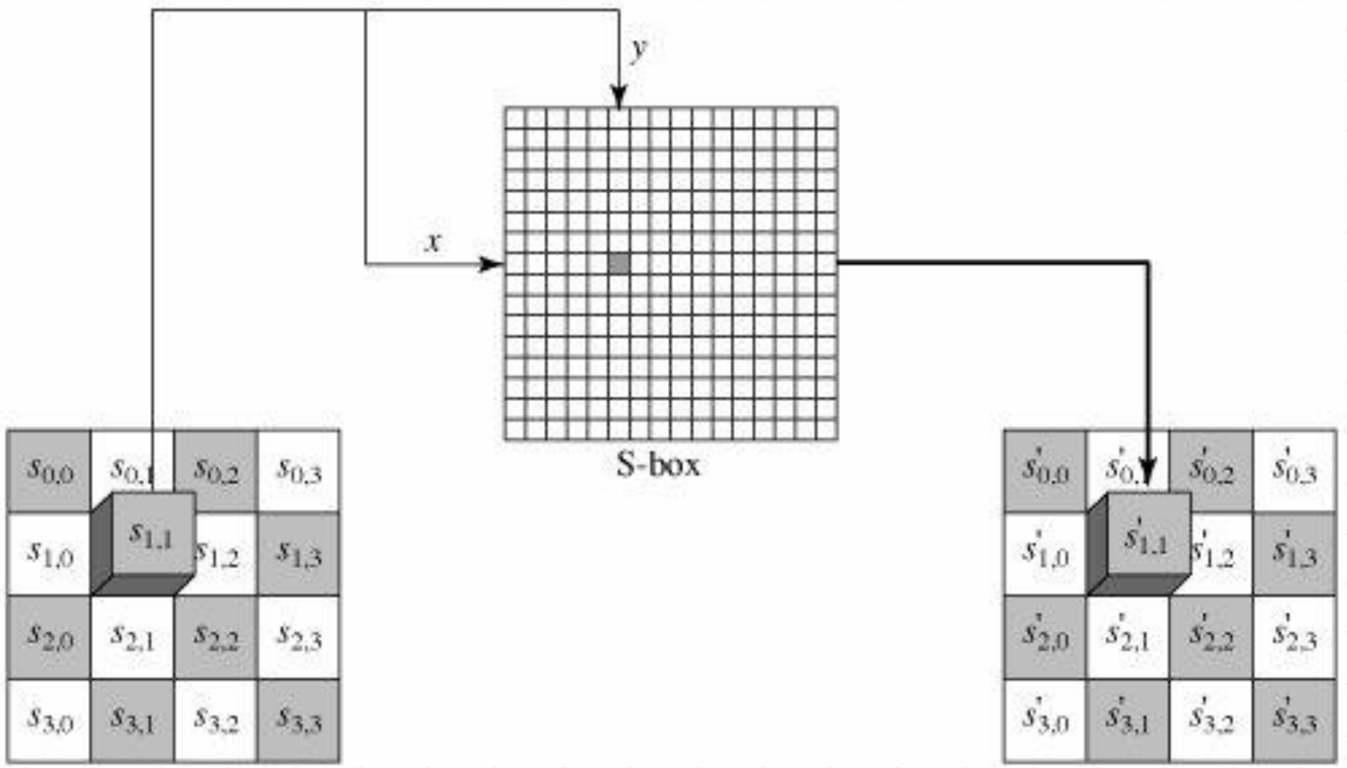
Substitute Bytes

- Byte-by-byte substitution
- Size of the lookup table is 16 × 16 (S-Box)
- To find the substitute byte for a given input byte;
- Divide the input byte into two 4-bit patterns, 0-15 (0 through F)
- One of the hex values is used as a row index and the other as a column index for reaching into the 16×16 lookup table

- Substitution byte (for each input byte) is found by using the same lookup table

Advanced Encryption Standard

Substitute Bytes



Advanced Encryption Standard

Substitute Bytes

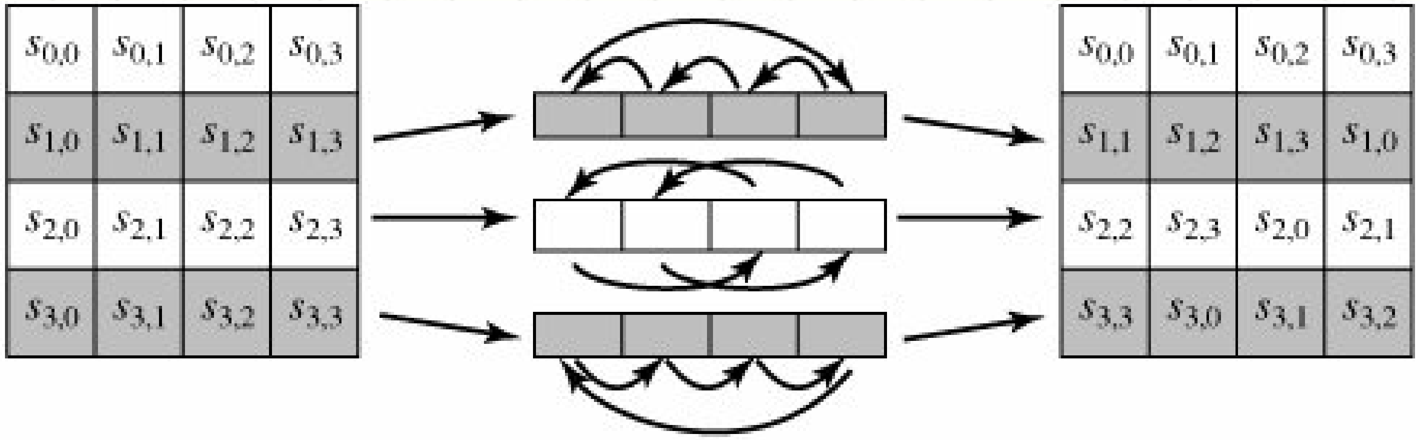
right (low-order) nibble

left (high-order) nibble	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	63	7c	77	7b	f2	6b	6f	c5	30	01	67	2b	fe	d7	ab	76
1	ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	c0
2	b7	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
3	04	c7	23	c3	18	96	05	9a	07	12	80	e2	eb	27	b2	75
4	09	83	2c	1a	1b	6e	5a	a0	52	3b	d6	b3	29	e3	2f	84
5	53	d1	00	ed	20	fc	b1	5b	6a	cb	be	39	4a	4c	58	cf
6	d0	ef	aa	fb	43	4d	33	85	45	f9	02	7f	50	3c	9f	a8
7	51	a3	40	8f	92	9d	38	f5	bc	b6	da	21	10	ff	f3	d2
8	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
9	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
a	e0	32	3a	0a	49	06	24	5c	c2	d3	ac	62	91	95	e4	79
b	e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	08
c	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
d	70	3e	b5	66	48	03	f6	0e	61	35	57	b9	86	c1	1d	9e
e	e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
f	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

Advanced Encryption Standard

Shift Rows

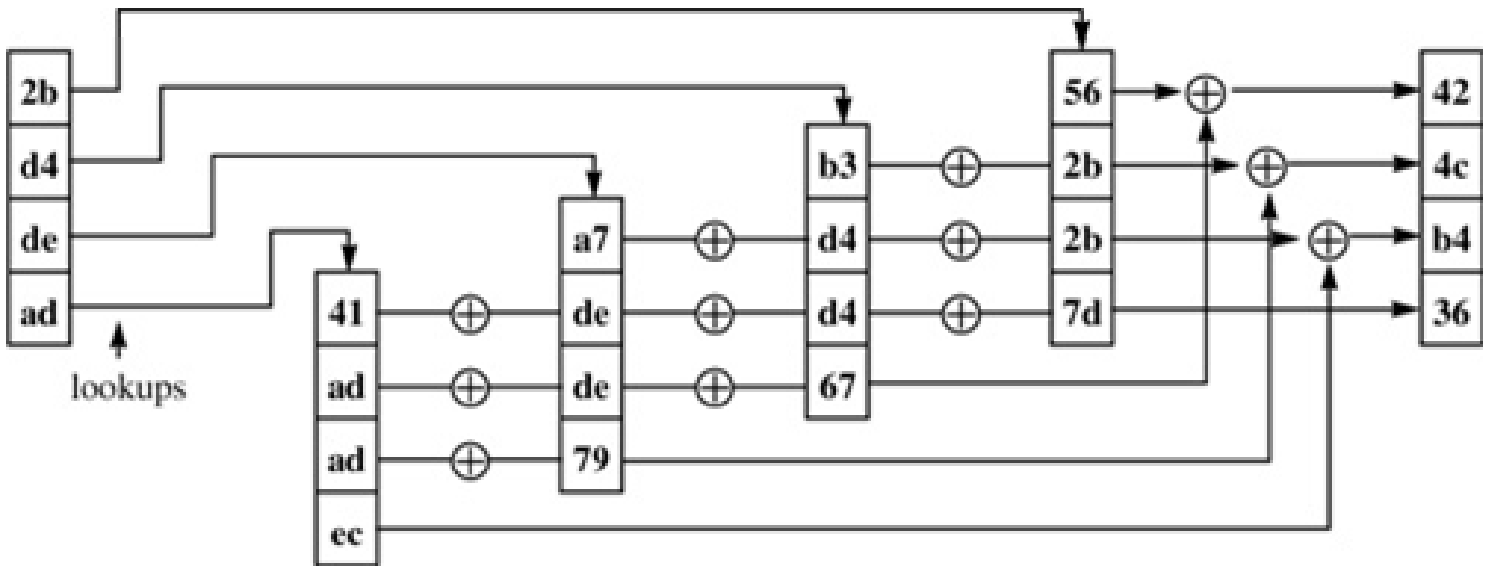
- Rotation or transformation consists of;
  - not shifting the first row of the state array at all
  - circularly shifting the second row by one byte to the left
  - circularly shifting the third row by two bytes to the left
  - circularly shifting the last row by three bytes to the left



Advanced Encryption Standard

Mix Columns

- This step replaces each byte of a column by a function of all the bytes in the same column



Advanced Encryption Standard

Mix Columns

- Lookup Table

right (low-order) nibble

left (high-order) nibble	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	00	02	04	06	08	0a	0c	0e	10	12	14	16	18	1a	1c	1e
	00	01	02	03	04	05	06	07	08	09	0a	0b	0c	0d	0e	0f
	00	01	02	03	04	05	06	07	08	09	0a	0b	0c	0d	0e	0f
	00	03	06	05	0c	0f	0a	09	18	1b	1e	1d	14	17	12	11
1	20	22	24	26	28	2a	2c	2e	30	32	34	36	38	3a	3c	3e
	10	11	12	13	14	15	16	17	18	19	1a	1b	1c	1d	1e	1f
	10	11	12	13	14	15	16	17	18	19	1a	1b	1c	1d	1e	1f
	30	33	36	35	3c	3f	3a	39	28	2b	2e	2d	24	27	22	21
2	40	42	44	46	48	4a	4c	4e	50	52	54	56	58	5a	5c	5e
	20	21	22	23	24	25	26	27	28	29	2a	2b	2c	2d	2e	2f
	20	21	22	23	24	25	26	27	28	29	2a	2b	2c	2d	2e	2f
	60	63	66	65	6c	6f	6a	69	78	7b	7e	7d	74	77	72	71
3	60	62	64	66	68	6a	6c	6e	70	72	74	76	78	7a	7c	7e
	30	31	32	33	34	35	36	37	38	39	3a	3b	3c	3d	3e	3f
	30	31	32	33	34	35	36	37	38	39	3a	3b	3c	3d	3e	3f
	50	53	56	55	5c	5f	5a	59	48	4b	4e	4d	44	47	42	41
4	80	82	84	86	88	8a	8c	8e	90	92	94	96	98	9a	9c	9e
	40	41	42	43	44	45	46	47	48	49	4a	4b	4c	4d	4e	4f
	40	41	42	43	44	45	46	47	48	49	4a	4b	4c	4d	4e	4f
	c0	c3	c6	c5	cc	cf	ca	c9	d8	db	de	dd	d4	d7	d2	d1
5	a0	a2	a4	a6	a8	aa	ac	ae	b0	b2	b4	b6	b8	ba	bc	be
	50	51	52	53	54	55	56	57	58	59	5a	5b	5c	5d	5e	5f
	50	51	52	53	54	55	56	57	58	59	5a	5b	5c	5d	5e	5f
	f0	f3	f6	f5	fc	ff	fa	f9	e8	eb	ee	ed	e4	e7	e2	e1
6	c0	c2	c4	c6	c8	ca	cc	ce	d0	d2	d4	d6	d8	da	dc	de
	60	61	62	63	64	65	66	67	68	69	6a	6b	6c	6d	6e	6f
	60	61	62	63	64	65	66	67	68	69	6a	6b	6c	6d	6e	6f
	a0	a3	a6	a5	ac	af	aa	a9	b8	bb	be	bd	b4	b7	b2	b1
7	e0	e2	e4	e6	e8	ea	ec	ee	f0	f2	f4	f6	f8	fa	fc	fe
	70	71	72	73	74	75	76	77	78	79	7a	7b	7c	7d	7e	7f
	70	71	72	73	74	75	76	77	78	79	7a	7b	7c	7d	7e	7f
	90	93	96	95	9c	9f	9a	99	88	8b	8e	8d	84	87	82	81
8	1b	19	1f	1d	13	11	17	15	0b	09	0f	0d	03	01	07	05
	80	81	82	83	84	85	86	87	88	89	8a	8b	8c	8d	8e	8f
	80	81	82	83	84	85	86	87	88	89	8a	8b	8c	8d	8e	8f
	9b	98	9d	9e	97	94	91	92	83	80	85	86	8f	8c	89	8a
9	3b	39	3f	3d	33	31	37	35	2b	29	2f	2d	23	21	27	25
	90	91	92	93	94	95	96	97	98	99	9a	9b	9c	9d	9e	9f
	90	91	92	93	94	95	96	97	98	99	9a	9b	9c	9d	9e	9f
	ab	a8	ad	ae	a7	a4	a1	a2	b3	b0	b5	b6	bf	bc	b9	ba
a	5b	59	5f	5d	53	51	57	55	4b	49	4f	4d	43	41	47	45
	a0	a1	a2	a3	a4	a5	a6	a7	a8	a9	aa	ab	ac	ad	ae	af
	a0	a1	a2	a3	a4	a5	a6	a7	a8	a9	aa	ab	ac	ad	ae	af
	fb	f8	fd	fe	f7	f4	f1	f2	e3	e0	e5	e6	ef	ec	e9	ea
b	7b	79	7f	7d	73	71	77	75	6b	69	6f	6d	63	61	67	65
	b0	b1	b2	b3	b4	b5	b6	b7	b8	b9	ba	bb	bc	bd	be	bf
	b0	b1	b2	b3	b4	b5	b6	b7	b8	b9	ba	bb	bc	bd	be	bf
	cb	c8	cd	ce	c7	c4	c1	c2	d3	d0	d5	d6	df	dc	d9	da
c	9b	99	9f	9d	93	91	97	95	8b	89	8f	8d	83	81	87	85
	c0	c1	c2	c3	c4	c5	c6	c7	c8	c9	ca	cb	cc	cd	ce	cf
	c0	c1	c2	c3	c4	c5	c6	c7	c8	c9	ca	cb	cc	cd	ce	cf
	5b	58	5d	5e	57	54	51	52	43	40	45	46	4f	4c	49	4a
d	bb	b9	bf	bd	b3	b1	b7	b5	ab	a9	af	ad	a3	a1	a7	a5
	d0	d1	d2	d3	d4	d5	d6	d7	d8	d9	da	db	dc	dd	de	df
	d0	d1	d2	d3	d4	d5	d6	d7	d8	d9	da	db	dc	dd	de	df
	6b	68	6d	6e	67	64	61	62	73	70	75	76	7f	7c	79	7a
e	db	d9	df	dd	d3	d1	d7	d5	cb	c9	cf	cd	c3	c1	c7	c5
	e0	e1	e2	e3	e4	e5	e6	e7	e8	e9	ea	eb	ec	ed	ee	ef
	e0	e1	e2	e3	e4	e5	e6	e7	e8	e9	ea	eb	ec	ed	ee	ef
	3b	38	3d	3e	37	34	31	32	23	20	25	26	2f	2c	29	2a
f	fb	f9	ff	fd	f3	f1	f7	f5	eb	e9	ef	ed	e3	e1	e7	e5
	f0	f1	f2	f3	f4	f5	f6	f7	f8	f9	fa	fb	fc	fd	fe	ff
	f0	f1	f2	f3	f4	f5	f6	f7	f8	f9	fa	fb	fc	fd	fe	ff
	0b	08	0d	0e	07	04	01	02	13	10	15	16	1f	1c	19	1a

Advanced Encryption Standard

Key Expansion

- 128 bits of the state array are bitwise XORed with the 128 bits of the round key
- AES Key Expansion algorithm is used to derive the 128-bit round key from the original 128-bit encryption key
- The algorithm first arranges the 16 bytes of the encryption key in the form of a 4 × 4 array of bytes

$$\begin{bmatrix} k_0 & k_4 & k_8 & k_{12} \\ k_1 & k_5 & k_9 & k_{13} \\ k_2 & k_6 & k_{10} & k_{14} \\ k_3 & k_7 & k_{11} & k_{15} \end{bmatrix}$$

Advanced Encryption Standard

Key Expansion

- First four bytes of the encryption key constitute the word w0, the next four bytes the word w1, and so on
- The algorithm subsequently expands the words [w0, w1, w2, w3] into a 44 word key schedule that can be labeled w0, w1, w2, w3, ..., w43

- Words [w0, w1, w2, w3] are bitwise XORed with the input block before the round-based processing begins
- The remaining 40 words are used four words at a time in each of the 10 rounds

Advanced Encryption Standard

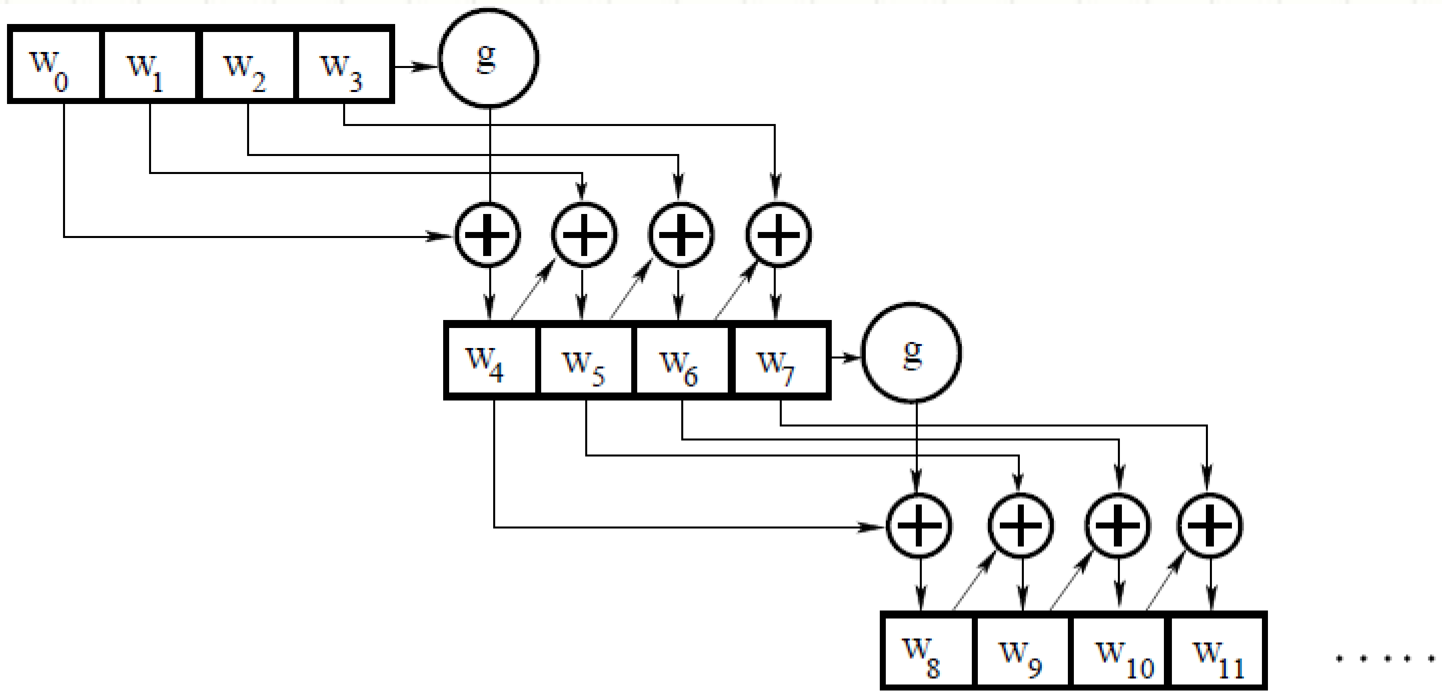
Key Expansion

- Now comes the interesting part;
- How does the Key Expansion Algorithm expand four words w0, w1, w2, w3 into the 44 words w0, w1, w2, w3, w4, w5, …., w43 ?

Advanced Encryption Standard

Key Expansion

- Key expansion takes place on a four-word to four-word basis
- Each grouping of four words decides what the next grouping of four words will be



Advanced Encryption Standard

Key Expansion

- Let's say that we have the four words of the round key for the ith round;

  w_i w_{i+1} w_{i+2} w_{i+3}

  - For these to serve as the round key for the ith round, i must be a multiple of 4
- For example, [w4, w5, w6, w7] is the round key for round 1, [w8, w9, w10, w11] round key for round 2, and so on

Advanced Encryption Standard

Key Expansion

- Now (for example) we need to determine the words [w_4 w_5 w_6 w_7] from the words [w_0 w_1 w_2 w_3]
- We have

$$w_5 = w_4 \otimes w_1$$
$$w_6 = w_5 \otimes w_2$$
$$w_7 = w_6 \otimes w_3$$

- Note that except for the first word in a new 4-word grouping, each word is an XOR of the previous word and the corresponding word in the previous 4-word grouping

Advanced Encryption Standard

Key Expansion

- So now we need to figure out w4 (or the beginning word of each 4-word grouping in the key expansion)
- It is obtained by:

$$w_4 = w_0 \otimes g(w_3)$$

- That is, XORing the first word of the last grouping with what is returned by applying a function g() to the last word of the previous 4-word grouping

Advanced Encryption Standard

Key Expansion

- The function g() consists of the following three steps:

1. Perform a one-byte circular rotation on the argument 4-byte word
2. Perform a byte substitution for each byte of the word returned by the previous step by using the same 16 × 16 lookup table as used in the SubBytes step of the encryption rounds
3. XOR the bytes obtained from the previous step with what is known as a round constant

Advanced Encryption Standard

Key Expansion

- The round constant is a word whose three rightmost bytes are always zero
- Therefore, XORing with the round constant amounts to XORing with just its leftmost byte
- The round constant for the ith round is denoted Rcon[i]

$$Rcon[i] = (RC[i], 0, 0, 0)$$

- The only non-zero byte in the round constants, RC[i], obeys the following recursion;

$$RC[1] = 1$$

$$RC[j] = 2 \times RC[j - 1]$$

* multiplication is defined over GF(2^8)

Advanced Encryption Standard

Key Expansion

- Lets see an example;
- The 4th column (w3) of the previous round key is rotated such that each element is moved up one row

2b	28	ab	09		cf
7e	ae	f7	cf	→	4f
15	d2	15	4f		3c
16	a6	88	3c		09

- Then put this result through a Sub Box, which replaces each 8 bits of the matrix with a corresponding 8-bit value from S-Box

cf		8a
4f	→	84
3c		eb
09		01

Advanced Encryption Standard

Key Expansion

- To generate the first column (w4) of the key, this result is XOR-ed with the first column of the key (w0) as well as a constant (Rcon) which is dependent on i

$$\begin{bmatrix} 2b \\ 7e \\ 15 \\ 16 \end{bmatrix} \oplus \begin{bmatrix} 8a \\ 84 \\ eb \\ 01 \end{bmatrix} \oplus \begin{bmatrix} 01 \\ 00 \\ 00 \\ 00 \end{bmatrix} = \begin{bmatrix} a0 \\ fa \\ fe \\ 17 \end{bmatrix}$$

- Values of Rcon based on i

01	02	04	08	10	20	40	80	1b	36

Advanced Encryption Standard

Key Expansion

- The second column (w5) is generated by XORing the 1st column (w4) of the key with the second column (w1) of the previous round key

$$\begin{bmatrix} 28 \\ ae \\ d2 \\ a6 \end{bmatrix} \oplus \begin{bmatrix} a0 \\ fa \\ fe \\ 17 \end{bmatrix} = \begin{bmatrix} 88 \\ 54 \\ 2c \\ b1 \end{bmatrix}$$

- This continues for the other two columns in order to generate the entire round key

$$\begin{bmatrix} 2b & 28 & ab & 09 & & \\ 7e & ae & f7 & cf & & \\ 15 & d2 & 15 & 4f & & \\ 16 & a6 & 88 & 3c & & \end{bmatrix} \rightarrow \begin{bmatrix} a0 & 88 & 23 & 2a \\ fa & 54 & a3 & 6c \\ fe & 2c & 39 & 76 \\ 17 & b1 & 39 & 05 \end{bmatrix}$$

Advanced Encryption Standard

<u>What makes AES a strong cipher</u>

- ❑ Round constant is for the purpose of destroying any symmetries that may have been introduced by other steps in the key expansion algorithm
- ❑ Note that if you change one bit of the encryption key, it will affect the round key for several rounds
- ❑ AES was published with a probability that <u>it will stay secure for at least 20 years</u>
- ❑ With 128 bit, ($2^{128} = 3.4 \times 10^{38}$) possible keys, a computer that tries 2^{55} keys per second needs 149 billion years to break AES

Advanced Encryption Standard

AES vs DES

- AES Key expansion algorithm ensures that AES has no weak keys
  - A weak key reduces the security of a cipher in a predictable manner
  - For example, DES is known to have weak keys (alternating ones and zeros), that produce identical round keys for each of the 16 rounds
- A weak key causes all round keys to become identical, which, in turn, causes the encryption to become self-inverting
  - That is, plain text encrypted and then encrypted again will lead back to the same plain text
- Since the small number of weak keys of DES are easily recognized, it is not considered to be a problem with that cipher

References

- AES official documentation

http://csrc.nist.gov/publications/fips/fips197/fips-197.pdf

- Rijndael Cipher Animation

http://www.codeplanet.eu/files/flash/Rijndael_Animation_v4_eng.swf

- Test AES encryption

https://www.openssl.org/

- Simplified AES

http://edipermadi.files.wordpress.com/2008/09/s-aes-spec.pdf

- "Cryptography and Network Security: Principles and Practice", 6th edition, by William Stallings
- "Network Security: Private Communication in a Public World", by Charlie Kaufman, Radia Perlman, Mike Speciner (chapter 3)

References

- Attacks on AES

State of AES attacks as of late 2010:

A. Kaminsky, M. Kurdziel, and S. Radziszowski. An overview of cryptanalysis research for the Advanced Encryption Standard. IEEE Military Communications Conference 2010 (MILCOM 2010), pages 1853-1859, San Jose, CA, USA, November 2010.

http://www.cs.rit.edu/~ark/20101102/milcom2010paper.pdf

http://www.cs.rit.edu/~ark/20101102/milcom2010v2.pdf

In August 2011, a key recovery attack (not a related key attack) on the full AES (not reduced-round AES) better than brute force (but just a little) was published:

A. Bogdanov, D. Khovratovich, and C. Rechberger. Biclique cryptanalysis of the full AES. Cryptology ePrint Archive, Report 2011/449, August 31, 2011.

http://eprint.iacr.org/2011/449

- Breaks AES-128 with $2^{126.1}$ work
- Breaks AES-192 with $2^{189.7}$ work
- Breaks AES-256 with $2^{254.4}$ work
- Also includes new breaks on reduced-round AES and on AES-based hash functions

AES is now (theoretically) broken!

THANKS